

Cybersickness Lies in the Eye of the Observer - Pupil Diameter as a Potential Indicator of Motion Sickness in Virtual Reality?

Katharina Pöhlmann
KITE Research Institute, UHN
Toronto, Canada

Aalim Makani
KITE Research Institute, UHN
Toronto, Canada

Raheleh Saryazdi
Trent University Durham
Oshawa, Canada

Behrang Keshavarz
KITE Research Institute, UHN
Toronto, Canada

ABSTRACT

Cybersickness is a widespread problem for many users of Virtual Reality systems. Changes in pupil diameter have been suggested as potential physiological correlates of cybersickness, but the relationship remains vague. Here, we further investigated how pupil diameter changes in relation to cybersickness by engaging participants in a passive locomotion through an outer-space environment. Participants who experienced sickness showed greater variance in pupil diameter compared to non-sick participants, whereas average pupil diameter did not differ. Our results suggest that irregular pupillary rhythms may be a potential correlate of cybersickness, which could be used to objectively identify cybersickness.

Index Terms: Human-centered computing—Human Computer Interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

Virtual Reality (VR) sickness remains one of the major barriers of the mainstream adoption of VR devices, severely affecting the experiences of its users and often resulting in early terminations of sessions as well as a reluctance to reengage with the medium. Symptoms of cybersickness are similar to those of traditional motion sickness, ranging from headaches and eye strain to dizziness and nausea. Its exact cause is not well understood, with the predominant theory involving a sensory conflict, suggesting that the discrepancy between perceived and expected visual, vestibular, and/or proprioceptive information causes these adverse symptoms.

To date, an objective measure that allows to reliably detect the presence and the severity of cybersickness does not exist. Instead, cybersickness is generally captured using subjective, self-report measures (e.g. [1, 2]), which are typically obtained after completing a VR session or during a session by interrupting the experience (potentially breaking immersion/presence). More recently, physiological measures have been investigated as alternative methods of assessing cybersickness objectively without interrupting the experience of the user and the results have been mixed [3]. If successful, these measures could potentially be used as a real-time feedback during VR experiences by applying mitigation techniques when needed, and ensuring a VR-sickness free user experience.

Many of the current headsets (Meta Quest Pro, Apple Vision Pro, etc.) have embedded sensors to capture physiological signals, such as eye tracking and face cameras, and heart rate sensors. Given their easy accessibility, physiological measures particularly eye-tracking are becoming a promising measure for detecting and predicting cybersickness. Although recent research has linked cybersickness to changes in certain oculomotor features such as increased pupil diameter and irregular patterns of pupillary rhythms [4, 7], no robust relationship between these features has yet been established. Therefore, the aim of the present work was to further investigate the relationship between changes in pupil diameter and cybersickness.



Figure 1: Example of the visual landscape (in the avatar condition) including examples of asteroids

2 METHOD

2.1 Participants

A total of 54 healthy volunteers participated in this study. Of those, 6 participants were excluded from the data analyses due technical issues with the eye tracker ($n=1$), increased nausea symptoms prior to performing the VR task ($n=2$), and insufficient pupillometry data due to early termination of the VR session ($n=3$). This resulted in a final sample size of $N = 48$ participants (age range 18-39 years; $M = 25.52$, $SD = 5.18$) with 26 self-identifying as female and 22 as male.

2.2 Measures

Pupil Diameter changes throughout the entirety of the study (measured in mm) were recorded using the build-in eye tracker (Tobii) of the HP Omnicept Reverb G2 headset. Changes in pupil diameter were reported at a sampling rate of 120Hz for both eyes separately and down-sampled to 1Hz for further analyses [7]. As part of the pre-processing of the raw data, pupil-size data points within a time window of 10ms pre and post blinks as well as data points that deviated more than ± 3 SDs from the mean pupil size were removed from the data analysis [6]. The pupil diameter for both eyes were then averaged, resulting in a single measures.

Cybersickness was rated by participants using both the Simulator Sickness Questionnaire (SSQ, [1]; pre and post VR session) as well as the Fast Motion Sickness Scale (FMS, [2]; administered every minute throughout the VR session). However, no meaningful correlations between the SSQ/FMS scores and pupil diameter were found; thus, these results are not further presented and discussed here. Instead, participants had to indicate after the VR session whether or not they experienced cybersickness using a binary *yes-no* question. Based on this question, participants were then post hoc assigned to a sick ($n = 30$) and a non-sick group ($n = 18$), which were used to compare changes in pupil diameter and rhythms between the two cybersickness groups.

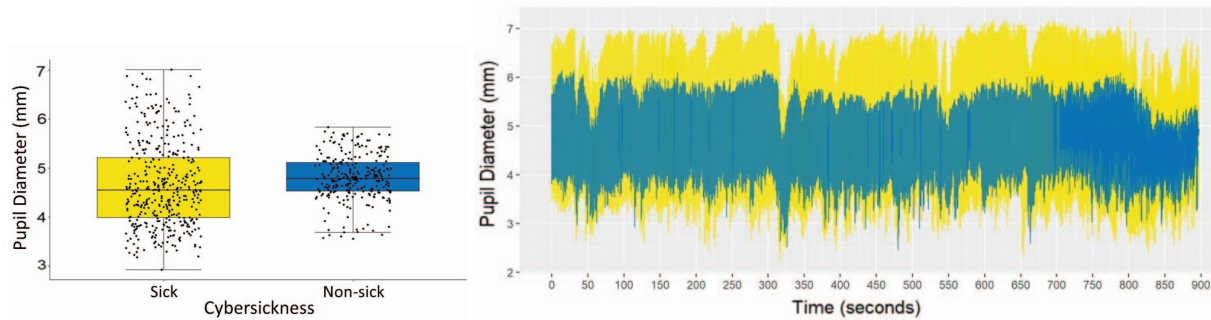


Figure 2: *Left*: Pupil Diameter in mm for the two cybersickness groups. Black lines represent the median and the colour of the boxes represents the cybersickness groups; *Right*: Pupil Diameter over the 15 min condition period for VR sick (yellow) and non-sick (blue) participants

2.3 Study Design and Procedure

During VR exposure, participants were passively moved through a virtual space scene following a pre-defined path for 15 minutes or until they reported moderate-to-high cybersickness (FMS > 10). This path included linear forward motion (with acceleration/deceleration) combined with rotations about the yaw, pitch, and roll axes (1). During the VR session, participants were asked to touch asteroid-like objects passing them in the virtual scene with their hands as quickly as possible to *destroy* them. Half of the participants completed the VR task either with an astronaut avatar present (i.e., visual representation of their hands and body) or absent (i.e., no visual representation of their hands or body). The avatar was used to manipulate the feeling of presence in the VR scene; however, because this manipulation is not relevant to the scope of the present paper, it will not be further discussed (see [5] for the results on presence and cybersickness). Thus, the pupil size data analyses was performed collapsing across both avatar conditions.

3 RESULT AND DISCUSSION

3.1 Difference in Pupil Diameter between Cybersickness groups

A linear mixed effect model was used for the comparison of average pupil diameter between the two cybersickness groups (contrast coded as no-sick = -1, sick = 1), including sickness group as fixed and participant as random effects. There was no significant difference in pupil diameter between the cybersickness groups, Estimate = -0.09, SE = 0.11, $t = -.08$, $p = .427$. Mean pupil diameter was similar in the sick ($M = 4.67$, $SD = 0.93$) and the non-sick group ($M = 4.82$, $SD = 0.55$), see Figure 2 left.

3.2 Differences in Pupillary Rhythm between Cybersickness groups

The increased standard deviation in the sick compared to non-sick group suggests a greater variance in pupil diameter in the sick group. To test whether this difference was statistically significant, two generalised least square models with cybersickness group as fixed effect and participant as random effect (assuming an autoregressive pattern in the pupil diameter by participant) were fitted and compared.

The model, which allowed for differing variance between the cybersickness groups (AIC = 205) provides a better fit compared to the simpler model (AIC = 217, $p < .001$), suggesting a significant difference in variance between the two groups. These findings indicate irregular pupillary rhythms in the sick group compared to the non-sick group during VR exposure.

3.3 Discussion

In sum, no relationship was found between overall pupil diameter and cybersickness, with mean pupil diameter being similar between

sick and non-sick participants. However, pupillary rhythms seem to differ greatly between the two groups, suggesting that the experience of cybersickness might be reflected by fluctuations in pupil diameter rather than the overall pupil size. Interestingly, this strong fluctuation in pupil diameter for individuals with cybersickness was found right from the beginning of the VR session (see 2 right), even though cybersickness gradually built up over the course of the VR session. This suggests that irregular rhythms in pupillary diameter might not necessarily reflect the severity of cybersickness, but rather differentiate more generally between sick and non-sick participants. As such, differences in pupil diameter rhythms might be a potentially useful predictor of cybersickness rather than a real-time measure. At this point, this explanation remains speculative and future research is needed to further test this hypothesis.

4 CONCLUSION

No difference in pupil diameter was found between participants who did or did not experience cybersickness, contradicting previous findings. However, our results demonstrated a relationship between pupil diameter rhythms and cybersickness. This work can build the foundation for future work looking further into pupil diameter rhythms as a potential objective marker of cybersickness (in combination with additional physiological/behavioural measures) and as a potential indicator of cybersickness susceptibility.

REFERENCES

- [1] R. S. Kennedy, N. E. Lane, K. S. Berbaum, and M. G. Lilienthal. Simulator sickness questionnaire: An enhanced method for quantifying simulator sickness. *The international journal of aviation psychology*, 3(3):203–220, 1993.
- [2] B. Keshavarz and H. Hecht. Validating an efficient method to quantify motion sickness. *Human factors*, 53(4):415–426, 2011.
- [3] B. Keshavarz, K. Peck, S. Rezaei, and B. Taati. Detecting and predicting visually induced motion sickness with physiological measures in combination with machine learning techniques. *International Journal of Psychophysiology*, 176:14–26, 2022.
- [4] P. Lopes, N. Tian, and R. Boulic. Eye thought you were sick! exploring eye behaviors for cybersickness detection in vr. In *Proceedings of the 13th ACM SIGGRAPH Conference on Motion, Interaction and Games*, pp. 1–10, 2020.
- [5] A. Makani, R. Saryazdi, S. Givertash, and B. Keshavarz. The presence of an avatar can reduce cybersickness in virtual reality. *Virtual Reality*, under review.
- [6] S. Mathôt and A. Vilotijević. Methods in cognitive pupillometry: Design, preprocessing, and statistical analysis. *Behavior Research Methods*, 55(6):3055–3077, 2023.
- [7] S. Park, S. Mun, J. Ha, and L. Kim. Non-contact measurement of motion sickness using pupillary rhythms from an infrared camera. *Sensors*, 21(14):4642, 2021.